

Rahul Sharma

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Education

George Mason University - College of Engineering & Computing
Bachelor of Science in Computer Science, GPA: 3.75

Aug 2022 – Dec 2026
Fairfax, VA

Experience

Software Engineer Intern — Rise Consultancy Edu, Remote

May 2026 – Jul 2026

- Built a React/TypeScript expenditure chart for a client dashboard in a production full-stack app with payment integrations.
- Fixed misleading spending trends by tracing missing months to sparse API data; normalized records to monthly intervals and filled gaps with zeros before rendering.
- Evaluated charting libraries against the design system; shipped through agile Jira sprints and senior engineers' PR reviews from staging to production.

Undergraduate Teaching Assistant

May 2026 – Present

George Mason University - Department of Computer Science

Fairfax, VA

- CS 310 (Data Structures), Summer 2026: supported 100+ students in two sections through weekly office hours and Piazza; helped debug Java lists, trees, hash tables, and graphs.
- CS 405 (Ethics and Law in Computing), Fall 2026: grade essays and mock-trial assignments; score the in-class trial.

Founding Engineer — TheCollegeTech - Educational Platform Solutions

Jun 2025 – Aug 2025

- Built a Learning Management System with TypeScript/React and Django REST, including authentication, account management, and role-based access control on Supabase.
- Designed course creation, enrollment, and content workflows with paginated and filtered SQL queries.
- Set up CI/CD with automated tests, wrote the API documentation, and ran code reviews.

College of Engineering IT Technician

Jun 2023 – Present

George Mason University - College of Engineering & Computing

Fairfax, VA

- Maintain and monitor 10–15 labs (~30 PCs each, ~95% uptime) with automated imaging, troubleshooting, and software tuning.
- Automate Windows, macOS, and Linux deployment and configuration with Python, Bash, and PowerShell scripts.
- Support faculty, staff, and graduate assistants; resolve production issues and document fixes in guides and tickets.

Projects

Drive | Go, React, TypeScript, PostgreSQL, S3 Multipart, Docker, Playwright

Jul 2026 – Sep 2026

- Built a self-hosted file storage platform with one Go binary serving the REST API and React/TypeScript app; uploads go directly from browser to storage over presigned S3 multipart URLs.
- Designed a fault-tolerant, resumable upload protocol with confirmed parts stored in PostgreSQL; transfers resume after dropped connections, closed tabs, or server crashes without resending confirmed parts.
- Verified integrity with per-part MD5/ETag checks and ledger reconciliation; tested an 11 GiB upload across 1,127 parts locally and a 220 MiB production round trip with SHA-256 verification.
- Deployed to production on Railway with Cloudflare R2 object storage; added Google OAuth, Argon2id auth, rate limits, quotas, and hourly garbage collection; built revocable share links with optional passwords, expiry, and per-session download caps, storing only SHA-256 token hashes.

XPen\$e | Next.js, Supabase, NFC, Auth0, Google Maps API, Perplexity AI API

Dec 2024

- Built a Next.js smart-wallet prototype that reads merchant data over NFC and suggests the best-reward card.
- Resolved merchant details with the Google Maps API and generated reward recommendations with the Perplexity AI API.

Technical Skills

Languages: Python, Go, TypeScript, JavaScript, Java, C++, SQL, Bash, PowerShell

Frontend: React, Next.js, Tailwind CSS

Backend: RESTful APIs, FastAPI, Django REST Framework, Node.js

Data & Infra: AWS S3, Cloudflare R2, PostgreSQL, Redis, Docker, Linux, Git, CI/CD

Testing: Playwright, Vitest, Go test